Ames Housing Price Prediction

Dataset: 2,930 rows and 79 predictors.
Target: SalePrice; trained on log1p(SalePrice) and converted back to dollars for metrics.

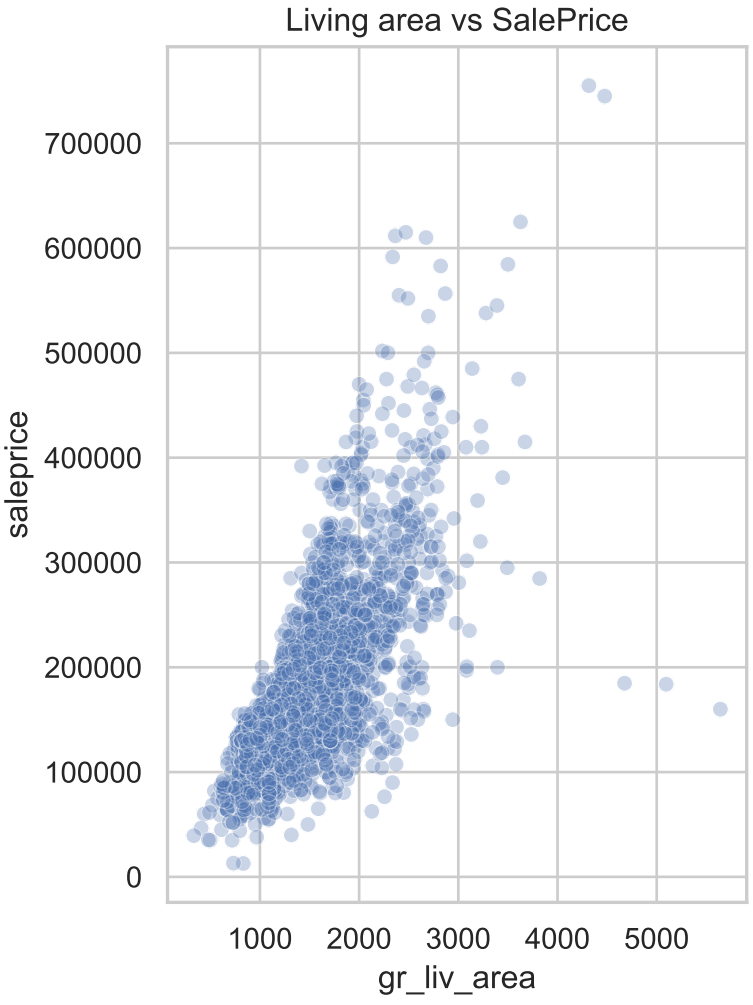
Preprocessing
- Median imputation and StandardScaler for numeric features.
- One-hot encoding for nominal categories.
- Explicit ordinal encoding for ordered quality/rating categories.
- All transformations are inside ColumnTransformer pipelines.

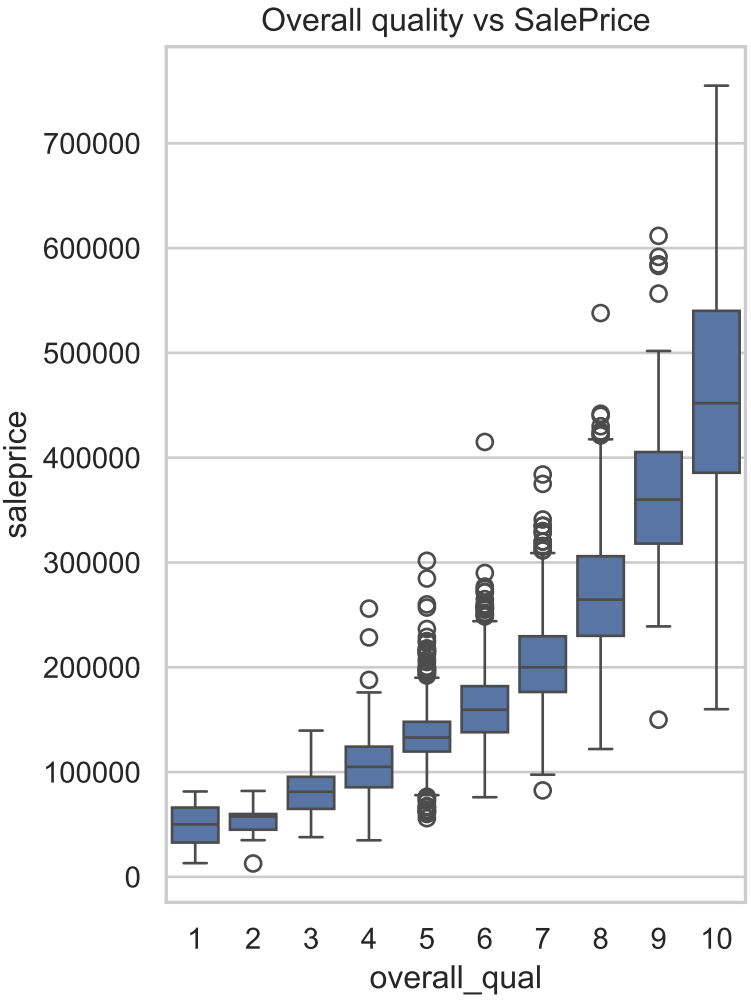
Model comparison

model	split	MAE	MSE	RMSE	R2
SVR	Train	10664.6	184884501.4	13597.2	1.0
SVR	Validation	14591.9	450167534.9	21217.2	0.9
SVR	Test	15054.9	834594431.9	28889.3	0.9
Ridge	Train	13786.9	546289780.6	23372.8	0.9
Ridge	Validation	14394.8	448603927.9	21180.3	0.9
Ridge	Test	16309.2	1241162792.3	35230.1	0.8

Recommendation
Ridge is selected using validation RMSE. Compare train versus validation RMSE for overfitting; the held-out test row is the final unbiased estimate.

EDA summary
SalePrice is right-skewed; living area and overall quality show strong positive relationships with price. Outliers and missingness were audited manually and with ydata-profiling.


Living area vs SalePrice
saleprice
0
100000
200000
300000
400000
500000
600000
700000
1000
2000
3000
4000
5000
gr_liv_area


Overall quality vs SalePrice
saleprice
0
100000
200000
300000
400000
500000
600000
700000
1
2
3
4
5
6
7
8
9
10
overall_qual